# Useful Memory Has a Horizon

Temporal Validity, Roughness, and Structural Limits for Finite-Memory Tracking under Drift

Vinicius Parreira

*Independent Researcher*

figueiredo@pm.me

May 2026

**Abstract**

In stationary settings, the law of large numbers makes memory an unambiguous statistical resource: more data improves estimation because the target distribution does not change over time. Under drift, this premise fails. Past evidence is generated by distributions that may no longer represent the present, so memory reduces variance while accumulating temporal misalignment. Once drift-induced misalignment dominates variance reduction, additional memory increases tracking error relative to the current distribution. Finite-memory tracking is therefore governed by a temporal-validity problem: how long can retained evidence continue to improve estimation of the present?

In the low-dimensional carrier regime the finite-sample term is modeled as a carrier law $C_K n^{-a}$, and the main text specializes to the root-$n$ case $a = 1/2$. For drift path classes in which temporal misalignment grows as $\zeta n^H$, this induces a roughness-indexed useful-memory horizon $n_H^* \asymp (C_K/\zeta)^{2/(1+2H)}$ together with the corresponding optimized upper-law scales. The worst-case $W_2$-Lipschitz envelope is the $H = 1$ member, recovering the cube-root horizon $n^* \asymp (C_K/\zeta)^{2/3}$. A Gaussian location lower bound shows that the same cube-root scaling defines a structural finite-memory floor in the worst-case Lipschitz regime, and the optimized ramp witness substantially strengthens the explicit lower-law constant while remaining subclass based.

This framework exposes a fundamental invalidity gap in non-stationary systems: retained evidence can lose temporal validity before a changepoint becomes statistically detectable. In finite samples, this structure appears through a finite optimum in horizon sweeps, roughness-dependent scaling across path classes, and measurable excess error when the operating horizon departs from the useful-memory scale.

**Keywords:** tracking under drift; useful memory; temporal validity; roughness-indexed horizon law; invalidity gap; Wasserstein distance.
**MSC 2020:** 93D05, 93D30, 68P99.

# 1 Introduction

In stationary settings, memory is an unambiguous statistical resource: more data reduces variance because the target distribution does not move. Under drift, that premise fails. In streaming estimation, monitoring, and online decision-making, the data-generating distribution itself changes over time, so a learner with finite memory faces a different question: how much of the past should still count once the world that produced it has started to move away?

That question creates a structural trade-off. More past data reduces sampling noise, but it also increases the age of the evidence used to represent the present. Under drift, memory is simultaneously helpful and harmful: short windows are fresh but noisy, while long windows are stable but stale. The paper studies this tension as a horizon problem for temporal validity, and the main object is a roughness-indexed family of horizon laws.

In the low-dimensional carrier regime the finite-sample term is modeled by a carrier law $C_K n^{-a}$, and the main text specializes to the root-$n$ case $a = 1/2$. In that regime, the theorem package begins with a roughness-indexed trade-off between finite-sample variance and drift-induced staleness. That family includes the worst-case $W_2$-Lipschitz cube-root law as its $H = 1$ member, recovers the intermediate $H = 1/2$ case inside the same object, and, in the worst-case Lipschitz regime, admits a Gaussian location lower bound establishing the same structural finite-memory floor. The optimized ramp witness materially improves the explicit constant available from that subproblem, even though the bound remains subclass based rather than constant-sharp. That floor is imposed by the changing path of distributions itself, not by inefficiency in a particular estimator.

For deterministic Holder-type path classes, temporal roughness changes how staleness accumulates, and the horizon law becomes a family indexed by roughness. The cube-root law is therefore the $H = 1$ member of the family rather than the whole theory.

The practical conflict is between two clocks. One clock measures the age at which retained evidence stops being valid for the present. The other measures when accumulated data provide enough statistical evidence for a changepoint alarm. Those clocks need not move together. The central conceptual distinction is therefore simple: temporal validity is not changepoint evidence. A detector can remain statistically silent while retained memory is already stale for the present.

The contributions are fivefold:

- The analysis formalizes useful memory under drift as a variance-staleness trade-off in Wasserstein geometry, with the carrier law written abstractly as $C_K n^{-a}$ and the main text specializing to the root-$n$ case $a = 1/2$.
- It proves a Holder-type roughness-indexed family of useful-memory horizon laws, with the worst-case $W_2$-Lipschitz cube-root regime recovered as the $H = 1$ member.
- It makes key special cases explicit inside the same family, including $H = 1$ and $H = 1/2$.
- It shows a Gaussian-location lower bound establishing a structural tracking floor in the critical-window regime, so the horizon is a property of the drift

geometry rather than an estimator-specific failure; the lower-bound claim is subclass based, exponent level, and supported by an optimized ramp witness with a substantially stronger explicit constant than the conservative initial construction.

- It identifies detector-silent staleness and measures the invalidity gap between temporal validity and changepoint evidence.

The empirical evidence follows that logic in stages. It first recovers the predicted $U$-curve and the finite useful-memory optimum. It then shows the gap between temporal validity and changepoint evidence, the regime-dependent scaling across roughness classes, and the nonlinear cost of horizon misalignment.

The theoretical results are stated in $W_2$; the Sinkhorn divergence is the computational measurement layer used in the synthetic experiments. The window size result of Hinder et al. [2023] motivates the broader issue, but the object here is more specific: finite memory has a temporal validity horizon under drift, and temporal roughness determines how that horizon scales across path classes.

## 2 Temporal Validity Under Drift

Under drift, memory is both a statistical resource and a temporal liability. Retaining more data reduces finite-sample noise, but it also increases the age of the evidence used to represent the present. The structural question is therefore not whether memory helps in the abstract, but how long retained evidence can remain temporally valid before staleness overwhelms the remaining variance gain.

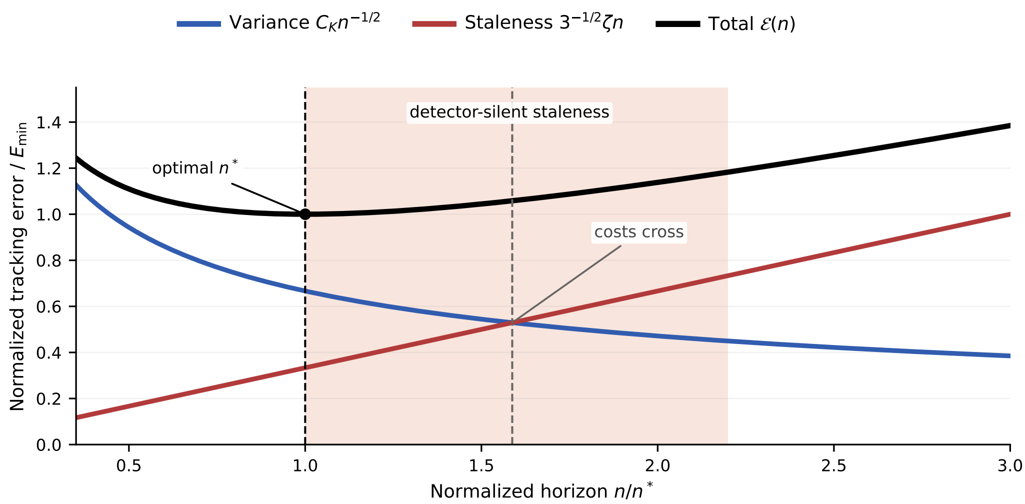


Figure 1: Two clocks of drift in normalized theorem units. The finite-sample term falls with horizon, the staleness term rises with age, and total error is minimized at the useful-memory horizon $n^*$. The shaded interval marks the invalidity gap: retained evidence can lose temporal validity before sufficient changepoint evidence accumulates.

For a finite-memory estimator with lag weights $w_0, \ldots, w_{n-1}$, the natural age variable is the effective lag

$$A(w) := \sum_{j=0}^{n-1} j\, w_j.$$

Its operational staleness at time $t$ is the weighted misalignment

$$S_t(w) := \sum_{j=0}^{n-1} w_j\, W_2\big(P^*_{t-j}, P^*_t\big)\,.$$

Under a local $W_2$-Lipschitz drift envelope with rate $\zeta_t$,

$$S_t(w) \le \zeta_t A(w).$$

For a uniform window, $A(w) = (n-1)/2$, so temporal age and window length are the same object up to constants. This is why horizon choice becomes the central statistical question.

At the theorem level, the structural object is the decomposition

$$\mathcal{E}(n) \le C_K n^{-a} + \text{Staleness}(n; \zeta, H),$$

where the first term is the finite-sample contribution and the second is the cost of retaining old evidence under drift. The main text specializes to the root-$n$ carrier case $a = 1/2$. The path class determines how the staleness term grows with lag, and therefore determines the useful-memory horizon.

## 2.1 Roughness-Indexed Horizon Family

The family of horizon laws comes first. The theorem-level object is a deterministic Holder-type envelope on paths of distributions: temporal roughness controls how staleness accumulates with lag, so different path classes induce different useful-memory scales. For $H \in (0, 1]$ and $\zeta > 0$, write

$$\mathfrak{P}_H(\zeta) := \big\{(P^*_s)_{s\in\mathbb{Z}} : W_2(P^*_{t-j}, P^*_t) \le \zeta j^H \text{ for all } j \ge 0\big\}$$

for the deterministic path class whose temporal misalignment is controlled by a one-sided Holder envelope at the present time $t$. The local Lipschitz regime is the special case $H = 1$.

**Lemma 2.1** (Holder staleness bound via averaged couplings)**.** *Let*

$$\bar{P}^{(n)}_t := \frac{1}{n}\sum_{j=0}^{n-1} P^*_{t-j}.$$

*If the target path belongs to $\mathfrak{P}_H(\zeta)$, then*

$$W_2^2\Big(\bar{P}^{(n)}_t, P^*_t\Big) \le \frac{\zeta^2}{n}\sum_{j=0}^{n-1} j^{2H},$$

*and therefore*

$$W_2\Big(\bar{P}^{(n)}_t, P^*_t\Big) \le c_H\, \zeta\, n^H$$

*for some constant $c_H > 0$ depending only on $H$.*

*Proof.* For each lag $j$, let $\pi_j$ be an optimal coupling between $P^*_{t-j}$ and $P^*_t$. Their average

$$\bar{\pi} := \frac{1}{n}\sum_{j=0}^{n-1} \pi_j$$

couples $\bar{P}_t^{(n)}$ with $P^*_t$, so

$$W_2^2\left(\bar{P}_t^{(n)}, P^*_t\right) \le \int \|x-y\|^2 \, d\bar{\pi}(x,y) = \frac{1}{n}\sum_{j=0}^{n-1} W_2^2\left(P^*_{t-j}, P^*_t\right).$$

Under the Holder envelope,

$$W_2^2\left(\bar{P}_t^{(n)}, P^*_t\right) \le \frac{\zeta^2}{n}\sum_{j=0}^{n-1} j^{2H}.$$

Since $n^{-1}\sum_{j=0}^{n-1} j^{2H} \asymp n^{2H}$ for fixed $H \in (0,1]$, taking square roots gives the stated bound. □

**Proposition 2.2** (Roughness-indexed horizon law)**.** *Let the target path belong to $\mathfrak{P}_H(\zeta)$ for some roughness index $H \in (0,1]$, so that the staleness growth obeys*

$$\mathrm{Staleness}_H(n;\zeta) \asymp \zeta n^H.$$

*If the finite-sample term remains of order $C_K n^{-1/2}$, then the tracking error bound*

$$\mathcal{E}_H(n) \le C_K n^{-1/2} + c_H\,\zeta n^H$$

*is minimized at*

$$n_H^* \asymp (C_K/\zeta)^{2/(1+2H)}, \qquad \mathcal{E}_{H,\min} \asymp C_K^{2H/(1+2H)}\,\zeta^{1/(1+2H)}.$$

*In particular, $H = 1$ recovers the cube-root law $n_1^* \asymp (C_K/\zeta)^{2/3}$, while $H = 1/2$ gives the intermediate special case $n_{1/2}^* \asymp C_K/\zeta$.*

*Proof.* By lemma 2.1, the deterministic staleness term is of order $\zeta n^H$. Balancing the decreasing statistical term against the increasing staleness term gives

$$C_K n^{-1/2} \asymp \zeta n^H,$$

which yields

$$n^{H+1/2} \asymp C_K/\zeta, \qquad n_H^* \asymp (C_K/\zeta)^{2/(1+2H)}.$$

Substituting this scale back into either term gives the stated minimum error. □

*Remark* 2.3 (Family interpretation). The point is not that the cube-root law is wrong. The point is that it is the $H = 1$ member of a broader roughness-indexed family. The $H = 1/2$ case is not a separate theorem story; it is simply an intermediate member recovered inside the same balance. As $H$ decreases, staleness accumulates more slowly with lag, so the useful-memory horizon moves accordingly.

## 2.2 Worst-Case Lipschitz Envelope

The worst-case Lipschitz regime is the linear-staleness envelope inside that family. It isolates the sharpest age penalty permitted by the local path bound.

**Theorem 2.4** (Lag-inclusive tracking error bound)**.** *Let $X_{t-j} \sim P^*_{t-j}$ and let*

$$\widehat{P}_t^{(n)} = \frac{1}{n}\sum_{j=0}^{n-1} \delta_{X_{t-j}}$$

*be the uniform-window empirical estimator. Assume the target path is locally $W_2$-Lipschitz with rate $\zeta$, so that*

$$W_2\big(P^*_{t-j}, P^*_t\big) \le \zeta j \qquad \text{for all } 0 \le j < n,$$

*and assume the finite-sample term obeys the carrier-regime bound*

$$\mathbb{E}\Big[W_2\Big(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)}\Big)\Big] \le C_K n^{-1/2},$$

*where $\bar{P}_t^{(n)} := n^{-1}\sum_{j=0}^{n-1} P^*_{t-j}$. Then the expected tracking error satisfies*

$$\mathcal{E}(n) \;\le\; C_K\, n^{-1/2} \;+\; 3^{-1/2}\,\zeta\, n,$$

*where the first term is the finite-sample statistical cost and the second term is the staleness cost induced by retaining old information.*

*Proof.* Apply the triangle inequality:

$$W_2\Big(\widehat{P}_t^{(n)}, P^*_t\Big) \le W_2\Big(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)}\Big) + W_2\Big(\bar{P}_t^{(n)}, P^*_t\Big).$$

The first term is controlled by the assumed finite-sample bound. For the second term, let $\pi_j$ be an optimal coupling between $P^*_{t-j}$ and $P^*_t$. The averaged coupling $\bar{\pi} := n^{-1}\sum_{j=0}^{n-1}\pi_j$ couples $\bar{P}_t^{(n)}$ with $P^*_t$, so the quadratic transport cost of that explicit coupling yields

$$W_2^2\Big(\bar{P}_t^{(n)}, P^*_t\Big) \le \int \|x-y\|^2\, d\bar{\pi}(x,y) = \frac{1}{n}\sum_{j=0}^{n-1} W_2^2\big(P^*_{t-j}, P^*_t\big).$$

Under the local Lipschitz envelope,

$$W_2^2\Big(\bar{P}_t^{(n)}, P^*_t\Big) \le \frac{\zeta^2}{n}\sum_{j=0}^{n-1} j^2 \le \frac{\zeta^2 n^2}{3},$$

so

$$W_2\Big(\bar{P}_t^{(n)}, P^*_t\Big) \le 3^{-1/2}\,\zeta n.$$

Combining the two bounds gives the claim. □

**Lemma 2.5** (I.i.d. reference for the carrier regime)**.** *Let $Q$ be a probability measure on $\mathbb{R}^{d_{\text{eff}}}$ with support contained in a ball of radius $R$ and finite second moment. Let $X_1, \ldots, X_n$ be i.i.d. samples from $Q$, and let $\hat{Q}_n := n^{-1}\sum_{i=1}^{n}\delta_{X_i}$. Then:*

- *If* $d_{\text{eff}} < 4$*, there exists a constant* $C_{d_{\text{eff}},R}$ *such that*

$$\mathbb{E}\, W_2(\hat{Q}_n, Q) \le C_{d_{\text{eff}},R}\, n^{-1/2}.$$

- *If* $d_{\text{eff}} = 4$*, the same bound holds with an additional* $(\log n)^{1/2}$ *factor.*

- *If* $d_{\text{eff}} > 4$*, the raw Wasserstein rate is slower than* $n^{-1/2}$*, so the constant-only form above should not be used.*

*In that high-dimensional regime, the statistical fluctuations do not follow the* $n^{-1/2}$ *carrier exponent, so the resulting horizon scaling lies outside the displayed cube-root regime. If the geometry is computed with debiased Sinkhorn divergence* $S_\varepsilon$ *on a bounded domain of diameter* $D$*, then for fixed* $\varepsilon > 0$ *the empirical fluctuations remain* $O(n^{-1/2})$ *with a constant* $L(D, \varepsilon)$ *that is dimension-free at fixed* $\varepsilon$ *and increases as* $\varepsilon \downarrow 0$*.*

*The low-dimensional* $W_2$ *threshold follows the standard empirical-measure rate theory of Fournier and Guillin [2015] and Niles-Weed and Bach [2019]. The fixed-*$\varepsilon$ *Sinkhorn statement is the corresponding regularized measurement-layer regime studied by Genevay et al. [2019] and, at the level of limit theorems, by Goldfeld et al. [2024].*

*A short numerical sanity check for this regime appears in Appendix A; it is not used as a theorem, but it matches the intended low-dimensional interpretation of the finite-sample term.*

*Remark* 2.6 (Carrier-regime status). Theorem 2.4 specializes to the $a = 1/2$ carrier case for the windowed triangular array $X_{t-j} \sim P^*_{t-j}$. Lemma 2.5 records the i.i.d. reference case that motivates this carrier exponent in the low-dimensional Wasserstein regime. Extending that rate from a fixed law $Q$ to the non-identically distributed window requires additional stability or coupling assumptions and is not claimed as a general theorem here. Recent concentration results for weighted empirical measures under Wasserstein drift and for triangular-array or dependent empirical measures in $W_1$ [Keehan et al., 2025, Amini and Vinas, 2026] indicate related concentration mechanisms under additional bounded-support or regularity assumptions, but they do not directly yield the present $W_2$ window-mixture statement, and we do not rely on such adaptations here.

The two terms in theorem 2.4 pull in opposite directions. Increasing $n$ reduces variance but raises the age of the retained evidence. That tension produces the worst-case Lipschitz horizon law.

**Proposition 2.7** (Optimal horizon in the Lipschitz case)**.** *Under theorem 2.4, the error* $\mathcal{E}(n)$ *is minimized at*

$$n^* \;=\; \left(\frac{\sqrt{3}\, C_K}{2\zeta}\right)^{2/3},$$

*with minimum value*

$$\mathcal{E}_{\min} \;=\; \frac{3^{5/6}}{2^{2/3}}\, C_K^{2/3}\, \zeta^{1/3}.$$

*Proof.* Differentiate the upper envelope $C_K n^{-1/2} + 3^{-1/2}\zeta n$ and solve

$$-\tfrac{1}{2}C_K n^{-3/2} + 3^{-1/2}\zeta = 0.$$

This gives $n^* = (\sqrt{3}C_K/(2\zeta))^{2/3}$. Substituting back yields the displayed minimum. □

*Remark* 2.8 (Scope of the horizon law). The cube-root law follows when the finite-sample term behaves as $\mathbb{E}\, W_2(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)}) \lesssim C_K n^{-1/2}$. More generally, if the finite-sample term admits scaling

$$\mathbb{E}\, W_2(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)}) \lesssim C_K n^{-a} \quad \text{for some } a > 0,$$

then balancing it against the linear staleness cost $c_{\text{lin}}\zeta n$ for some explicit constant $c_{\text{lin}} > 0$ yields

$$n^*(a) \;=\; \left(\frac{a\, C_K}{c_{\text{lin}}\zeta}\right)^{\frac{1}{a+1}}, \qquad \mathcal{E}_{\min}(a) \;=\; (a+1)\, a^{-a/(a+1)}\, c_{\text{lin}}^{a/(a+1)}\, C_K^{1/(a+1)}\, \zeta^{a/(a+1)}.$$

The main text uses only the $a = 1/2$ regime with $c_{\text{lin}} = 3^{-1/2}$. The displayed cube-root law is therefore the worst-case linear-staleness member of the broader family.

*Remark* 2.9 (Measurement layer). The theorem is stated in $W_2$, so $\varepsilon$ does not enter the drift law itself. In the synthetic experiments, the observable geometry is often a debiased Sinkhorn divergence with fixed $\varepsilon$, which keeps the same $n^{-1/2}$ carrier exponent while changing constants. The computational layer is therefore separate from the structural horizon law.

**Corollary 2.10** (EWMA analogue). *For exponential weights $w_j = \alpha(1-\alpha)^j$, the effective lag is $\tau_\alpha = (1-\alpha)/\alpha$ and the effective sample size scales as $n_{\text{eff}} \asymp 1/\alpha$. Balancing the same statistical and staleness terms therefore yields the same cube-root exponent in the effective memory scale. This is the exponentially weighted analogue of the $H = 1$ case.*

## 2.3 A Structural Finite-Memory Floor

The upper law is not merely the optimum of one estimator. The same scale appears in a lower-bound witness, which makes the useful-memory horizon structural for the stated class.

**Proposition 2.11** (Finite-memory floor via a Gaussian location witness). *Let $\mathcal{P}_{\zeta,m}$ be the class of length-$m$ distribution paths such that:*

1. *each path is $W_2$-Lipschitz with rate at most $\zeta$;*
2. *the lower-bound construction is carried out on the Gaussian location subclass $X_t \sim \mathcal{N}(\mu_t, \sigma^2 I_d)$ with $\|\mu_t - \mu_s\|_2 \le \zeta|t-s|$;*
3. *the estimator at time $m$ only uses the last $m$ samples.*

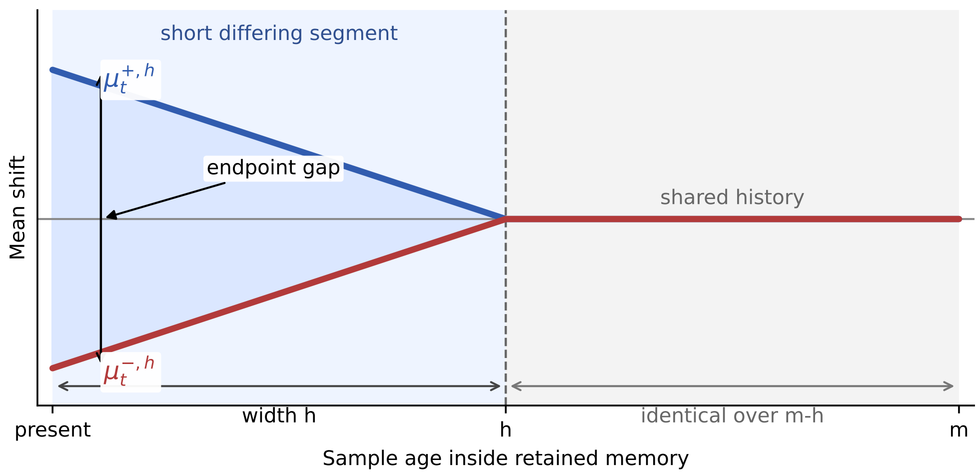


Figure 2: Lower-bound witness on the Gaussian location subclass. The two paths share most of the retained history and differ only on a short recent segment of width $h$, yet they remain separated at the present. Optimizing that recent ramp segment already yields a substantially stronger explicit witness constant at the same cube-root scale as the upper-law optimum.

*Measure risk by the endpoint $W_2$ error, which coincides with Euclidean endpoint error on the Gaussian location subclass. Then, in the critical-window regime, there exist explicit constants $c_0, c > 0$ such that the minimax tracking error over that class cannot beat the same exponent-level variance-plus-staleness scaling:*

$$\inf_{\hat{\mathcal{P}}_m} \sup_{\mathcal{P}^*} \mathbb{E}\, d\big(\hat{\mathcal{P}}_m, \mathcal{P}^*\big) \gtrsim \max\big\{c_0\, \sigma\, m^{-1/2},\ c\, \sigma^{2/3} \zeta^{1/3}\big\}\,.$$

*Consequently, the cube-root horizon is not merely the optimum of one estimator; it is the structural finite-memory floor on this class at the cube-root scale.*

**Proof.** Write $R_m^* := \inf_{\delta_m} R_m(\delta_m)$. We exhibit two independent obstructions. The first is the ordinary finite-sample estimation floor, obtained by restricting to the constant-path subclass $\mu_s \equiv \theta$, which is contained in the admissible drift class. For the two hypotheses $\theta_\pm = \pm\sigma/(2\sqrt{m})$, the induced product-Gaussian laws have Kullback–Leibler divergence

$$\mathrm{KL}(P_+, P_-) = \frac{1}{2\sigma^2} \sum_{j=1}^{m} (\theta_+ - \theta_-)^2 = \frac{1}{2\sigma^2} m \left(\frac{\sigma}{\sqrt{m}}\right)^2 = \frac{1}{2}.$$

For any estimator $\delta_m$, the plug-in test $\psi = \mathbf{1}\{\delta_m \geq 0\}$ must err whenever the estimate falls on the wrong side of the origin; on that event the endpoint error is at least $\sigma/(2\sqrt{m})$. By Le Cam's two-point lemma and Pinsker's inequality, the average testing error is at least

$$\frac{1 - \mathrm{TV}(P_+, P_-)}{2} \geq \frac{1 - \sqrt{\mathrm{KL}(P_+, P_-)/2}}{2} \geq \frac{1}{4}.$$

Hence

$$R_m^* \geq \frac{\sigma}{8\sqrt{m}},$$

so one valid explicit choice is $c_0 = 1/8$.

The second obstruction is local drift. Fix any $1 \le h \le m$ and define

$$\mu_{t-j}^{\pm,h} \coloneqq \pm\,\zeta\,(h-j)_+, \qquad 0 \le j \le m-1,$$

where $(u)_+ \coloneqq \max\{u, 0\}$. Each path belongs to the admissible class because its slope magnitude is exactly $\zeta$ on the active segment. The endpoint gap is

$$|\mu_t^{+,h} - \mu_t^{-,h}| = 2\zeta h.$$

The corresponding Gaussian product laws satisfy

$$\mathrm{KL}(P_{+,h}, P_{-,h}) = \frac{1}{2\sigma^2}\sum_{j=0}^{m-1}\bigl(2\zeta(h-j)_+\bigr)^2 = \frac{2\zeta^2}{\sigma^2}\sum_{r=1}^{h} r^2.$$

Applying the same reduction from estimation to testing, now with endpoint error threshold $\zeta h$, and using Le Cam together with Pinsker gives average testing error at least

$$\frac{1-\mathrm{TV}(P_{+,h}, P_{-,h})}{2} \ge \frac{1-\sqrt{\mathrm{KL}(P_{+,h}, P_{-,h})/2}}{2}$$

Therefore

$$R_m^* \;\ge\; \frac{\zeta h}{2}\left(1-\sqrt{\frac{\zeta^2}{\sigma^2}\sum_{r=1}^{h} r^2}\right),$$

which holds for every $1 \le h \le m$.

To optimize the same ramp witness in the critical-window regime, write $h_\star \coloneqq (\sigma/\zeta)^{2/3}$ and choose

$$a_\star \coloneqq \left(\frac{2\sqrt{3}}{5}\right)^{2/3}, \qquad \bar h \coloneqq \lfloor a_\star h_\star \rfloor.$$

When $1 \le \bar h \le m$, we have

$$\sum_{r=1}^{\bar h} r^2 = \frac{\bar h^3}{3} + O(\bar h^2),$$

so, since $\bar h = a_\star h_\star + O(1)$,

$$\mathrm{KL}(P_{+,\bar h}, P_{-,\bar h}) = \frac{2\zeta^2}{\sigma^2}\sum_{r=1}^{\bar h} r^2 = \frac{2a_\star^3}{3} + o(1) = \frac{8}{25} + o(1).$$

Hence

$$\frac{1-\mathrm{TV}(P_{+,\bar h}, P_{-,\bar h})}{2} \ge \frac{1-\sqrt{\mathrm{KL}(P_{+,\bar h}, P_{-,\bar h})/2}}{2} = \frac{3}{10} + o(1),$$

while the endpoint threshold obeys

$$\zeta \bar h = a_\star\,\sigma^{2/3}\zeta^{1/3} + o(\sigma^{2/3}\zeta^{1/3}).$$

Therefore the ramp witness gives

$$R_m^* \;\geq\; \left(\frac{3}{10}\,a_\star + o(1)\right)\sigma^{2/3}\zeta^{1/3} = \left(\frac{3}{10}\left(\frac{2\sqrt{3}}{5}\right)^{2/3} + o(1)\right)\sigma^{2/3}\zeta^{1/3}.$$

Taking the maximum of the constant-path and ramp bounds proves the displayed lower bound. In particular, the same ramp family already yields the explicit asymptotic constant

$$c = \frac{3}{10}\left(\frac{2\sqrt{3}}{5}\right)^{2/3} \approx 0.2349,$$

which is substantially larger than the more conservative witness constant used in earlier drafts, even though the result remains exponent-level rather than constant-sharp.

*Remark* 2.12 (Scope of the lower bound). The lower bound is subclass-based. It uses a Gaussian location witness whose mean path is itself $W_2$-Lipschitz, and the distance in the lower bound is the endpoint $W_2$ error, which becomes Euclidean on that subclass. That is enough to make the result minimax for any larger class containing the same witness, including the ambient $W_2$-Lipschitz path class. The proposition therefore does not claim that every path is Gaussian; it claims that the worst case already appears inside the Gaussian mean-shift subproblem.

At the Lipschitz endpoint $H = 1$, this witness matches the cube-root scaling of the upper law, so the lower bound is exponent-tight there. It is not intended to identify the exact hardest path for the full class, to be constant sharp, or to furnish matching lower bounds for the broader Holder family $H \in (0, 1)$.

# 3 Empirical Evidence

The experiments make the theory visible in finite samples. They do not compare named interventions. They read observed tracking behavior against the useful-memory scale implied by the regime under study. Throughout, the oracle reference is the horizon predicted by the same variance–staleness law that drives the theory section.

Four signatures matter. First, horizon sweeps should exhibit a finite optimum. Second, that optimum should scale differently across roughness classes. Third, the operating horizon can lose temporal validity before a detector accumulates enough evidence to signal change. Fourth, excess error should increase once the operating horizon departs materially from the useful-memory scale.

## 3.1 Structural U-Curve

The Gaussian sweep isolates the variance–staleness trade-off directly. Small windows are variance dominated, large windows are stale, and the optimum lies at an interior horizon rather than at either boundary.

This is the basic finite-memory signature. Under stationarity, more memory would continue to reduce variance. Under drift, the sweep bends upward once temporal misalignment overtakes the remaining variance gain.

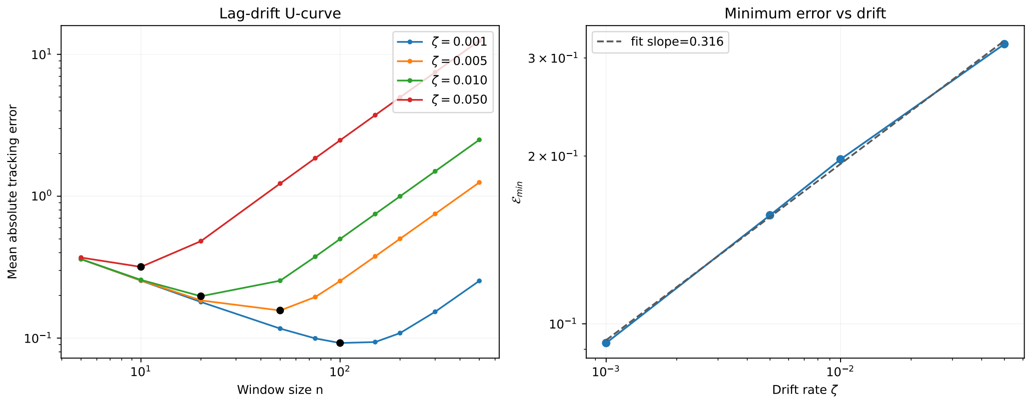


Figure 3: Structural U-curve for finite-memory tracking. Left: tracking error as a function of horizon for several drift strengths. Right: the minimum achievable error decreases as drift weakens. The optimum remains finite across regimes, which makes the useful-memory horizon visible before any path-class refinement.

## 3.2 Regime-Dependent Scaling Across Roughness Classes

The next experiment varies the roughness index directly. For each Holder-type regime, the construction uses a finite-sample surrogate whose error obeys the same variance–staleness decomposition as the theory,

$$\mathcal{E}(n) \approx \sigma n^{-1/2} + \zeta n^{H},$$

then measures the empirically optimal horizon across a sweep of $\zeta$ values.

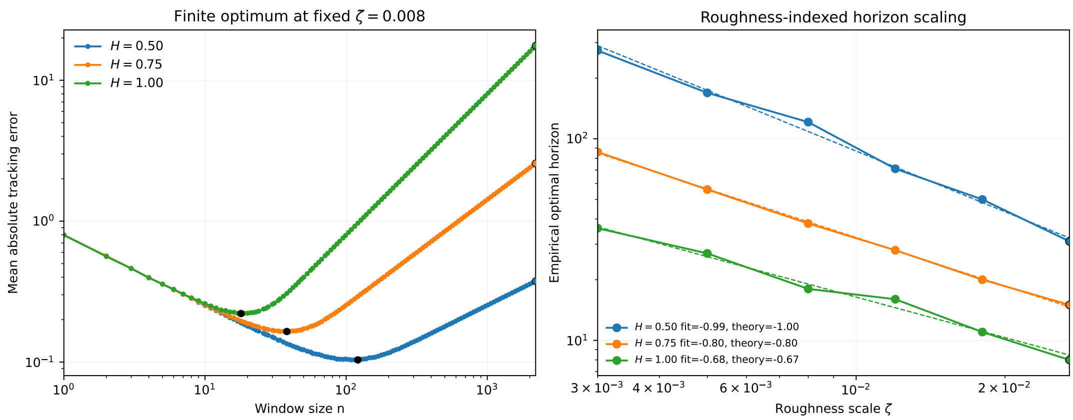


Figure 4: Roughness-dependent horizon scaling. Left: each roughness class still produces a finite optimum, but the location of that optimum shifts with $H$. Right: the empirical optimal horizon follows a distinct log–log scaling law in each class. The $H = 1$ curve recovers the worst-case Lipschitz member, the $H = 1/2$ curve gives the intermediate special case, and smaller $H$ values fall on the same roughness-indexed family rather than defining separate stories.

The point of this figure is structural rather than cosmetic. If the cube-root law were the whole story, the right panel would collapse to a single slope. It does not.

The empirical optima separate by roughness class, and the recovered log–log slopes follow the predicted roughness-indexed family.

## 3.3 Detector-Silent Staleness

Temporal validity and changepoint evidence are different clocks. To make that separation explicit, consider a stream whose local drift rate ramps upward over time. The useful-memory horizon $n_t^*$ contracts as the ramp steepens, while a long operating horizon remains fixed. Define

$$\Delta_{\text{inv}} := t_{\text{detect}} - t_{\text{valid}},$$

where $t_{\text{valid}}$ is the first time the operating horizon persistently exceeds the useful-memory scale, and $t_{\text{detect}}$ is the first detector alarm that follows. This separation is consistent with the basic logic of sequential changepoint tests: thresholds are tuned to suppress false alarms, so an integrated detector statistic typically requires additional accumulation time before crossing an alarm boundary [Page, 1954, Hinkley, 1970].

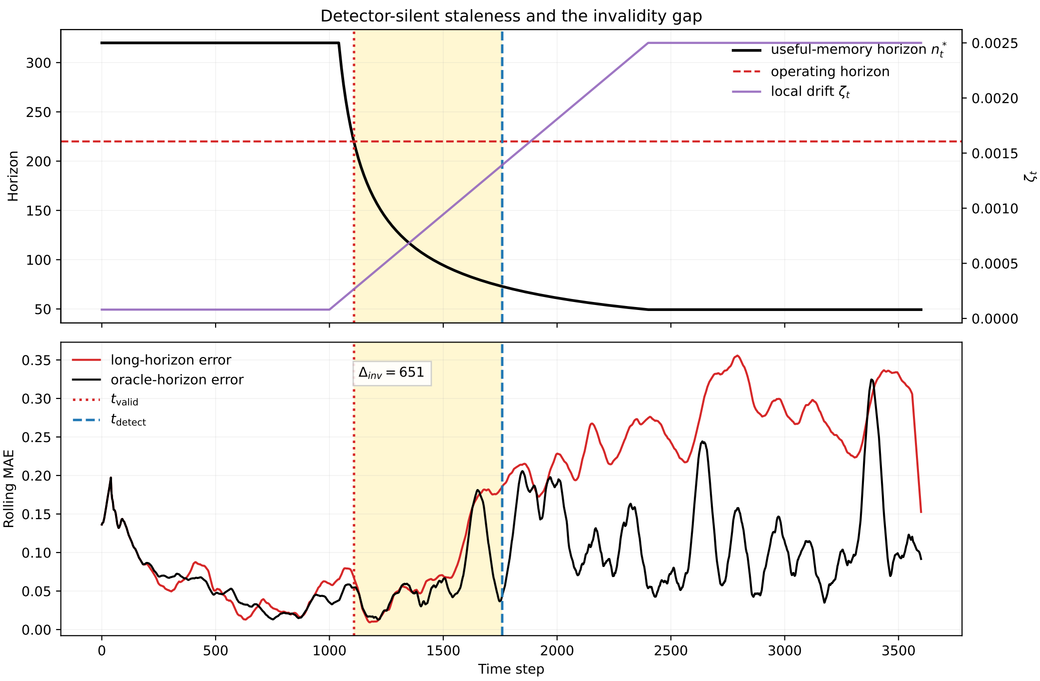


Figure 5: Detector-silent staleness. As drift intensifies, the useful-memory horizon contracts until the operating horizon becomes stale at $t_{\text{valid}}$; the detector responds only later at $t_{\text{detect}}$, leaving a strictly positive invalidity gap $\Delta_{\text{inv}}$. In the lower panel, stale memory already imposes excess tracking error before the alarm appears.

Across detector settings, the measured gap remains positive in every reported run. The detector-threshold ablation in table 1 shows mean gaps of roughly $5 \times 10^2$ steps over the tested settings. This is the observable form of detector-silent staleness: evidence can become obsolete while standard detection remains silent.

The phenomenon is not specific to ADWIN. The detector-family comparison in table 2 shows that the gap remains strictly positive for Page–Hinkley as well, even though the alarm timing changes.

Table 1: Detector-threshold ablation for the invalidity gap. Across the tested ADWIN thresholds, the mean detection time remains strictly later than the mean validity-expiry time, so the measured invalidity gap stays positive throughout.

| $\delta$ | $\bar{t}_{\text{valid}}$ | $\bar{t}_{\text{detect}}$ | $\overline{\Delta}_{\text{inv}}$ |
|---|---|---|---|
| 0.0005 | 1108.0 | 1671.0 | 563.0 |
| 0.0010 | 1108.0 | 1657.7 | 549.7 |
| 0.0020 | 1108.0 | 1641.7 | 533.7 |
| 0.0040 | 1108.0 | 1631.0 | 523.0 |

Table 2: Detector-family comparison for the invalidity gap. Using the same drift ramp and operating horizon, the positive gap persists for both ADWIN and Page–Hinkley, even though their changepoint clocks differ.

| Detector | $\bar{t}_{\text{valid}}$ | $\bar{t}_{\text{detect}}$ | $\overline{\Delta}_{\text{inv}}$ |
|---|---|---|---|
| ADWIN | 1108.0 | 1641.7 | 533.7 |
| Page–Hinkley | 1108.0 | 1410.6 | 302.6 |

## 3.4 Horizon Misalignment Cost

The final question is how sharply performance degrades when the operating horizon misses the useful-memory scale. Using the same roughness-indexed surrogate, fig. 6 plots excess error against the relative horizon gap $|\log(n/n^*)|$.

This is the finite-sample reason the horizon itself matters. The optimum is not only finite; it is operationally relevant. Errors remain modest in a narrow band around the useful-memory scale, then increase on either side as variance or staleness takes over.

## 3.5 What the Evidence Shows

Taken together, the experiments reveal four features that match the theory. First, finite-memory tracking exhibits a structural U-curve rather than a boundary optimum. Second, the useful-memory scale is roughness dependent, with the worst-case Lipschitz cube-root law appearing as the $H = 1$ member of a wider family. Third, temporal validity can fail before changepoint evidence appears, which produces a measurable invalidity gap. Fourth, missing the useful-memory scale carries a visible finite-sample cost even when the operating horizon is only moderately misaligned.

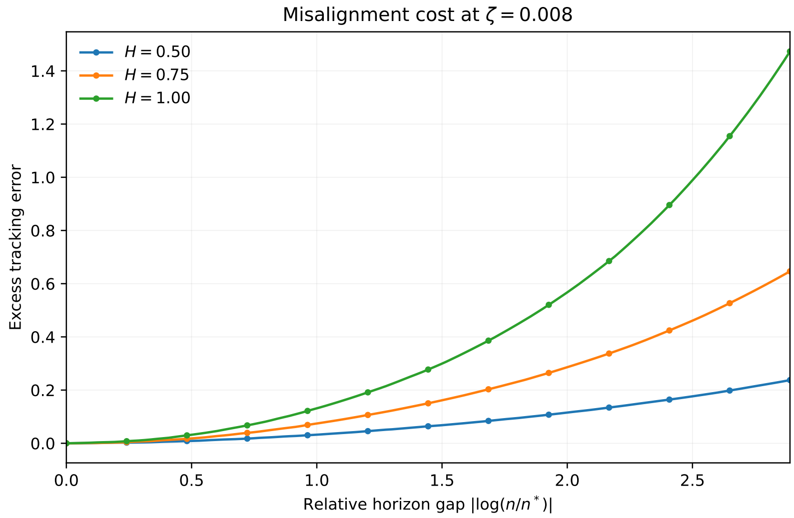


Figure 6: Cost of horizon misalignment. Excess tracking error is smallest near the useful-memory scale and rises as the operating horizon moves away from it. Being too short is noisy, being too long is stale, and the penalty steepens at class-dependent rates once the operating horizon leaves the low-cost band around $n^*$.

# 4 Related Work

**Nonstationary optimization and dynamic regret.** Dynamic-regret and variation-budget results show that under drift, window size or forgetting scale must depend on how quickly the target moves [Besbes et al., 2015, Cheung et al., 2019, Yuan and Lamperski, 2020, Mazzetto and Upfal, 2023, Wang et al., 2024]. Those works are close in spirit, but their primary object is regret under a moving comparator. Here the primary object is different: the temporal validity of retained evidence in a distribution-tracking problem. The horizon is not a tuning parameter attached to regret analysis; it is the statistical object being characterized.

**Adaptive windowing and drift detection.** Adaptive-window and concept-drift methods ask when accumulated evidence is sufficient to react [Bifet and Gavaldà, 2007, Gama et al., 2004, 2014]. Hinder et al. [2023] emphasize that the relevant window scale is itself part of the problem. The present question is adjacent but distinct. Temporal validity is not changepoint evidence: memory can stop representing the present before the available data support a detector alarm. The invalidity gap makes that distinction measurable rather than rhetorical.

**Adaptive filtering and state-space tracking.** Classical adaptive filtering and state-space tracking also balance smoothing against responsiveness by tuning forgetting factors, gain schedules, or process noise models [Kalman, 1960, Jazwinski, 1970]. Those analyses usually work under stochastic state-space assumptions and ask for Bayes-optimal tracking of a latent process. The present paper instead studies deterministic Wasserstein drift envelopes and characterizes a worst-case temporal-validity horizon.

**Locally stationary smoothing.** Locally stationary time-series analysis likewise studies how window or bandwidth choice must adapt to time variation [Dahlhaus, 1997]. That literature is close in its bias–variance logic, but the moving object there is typically a local spectrum or regression function. Here the moving object is a path of distributions, and the age penalty is measured directly in transport geometry.

**Distribution tracking under transport geometry.** Wasserstein distance and Sinkhorn-type estimators are natural when drift lives on paths of distributions rather than on scalar losses [Fournier and Guillin, 2015, Genevay et al., 2019, Niles-Weed and Bach, 2019, Goldfeld et al., 2024, Akyildiz et al., 2024]. They supply the geometric language in which staleness becomes a transport error between past and present. In this setting, the trade-off is not simply between adaptation and variance, but between finite-sample accuracy and temporal misalignment measured in distributional geometry.

**Wasserstein nonstationarity and robust optimization.** Jiang, Li, and Zhang study online stochastic optimization under a Wasserstein nonstationarity measure and obtain distribution-aware regret guarantees [Jiang et al., 2020]. Keehan, Anderson, and Wiesemann analyze related drift-aware weighting questions in Wasserstein distributionally robust optimization [Keehan et al., 2025]. Amini and Vinas give a recent moment-method route to triangular-array concentration in $W_1$ [Amini and Vinas, 2026]. Those results are adjacent to the carrier-regime issue studied here, but they do not by themselves close the present $W_2$ window-mixture concentration problem. The present focus is narrower and more structural: for finite-memory tracking, how long can retained evidence continue to improve estimation of the current distribution before age becomes a dominant error source?

**Freshness and Age of Information.** The Age-of-Information literature treats age as a resource constraint and asks how update policies balance communication cost against stale information [Kaul et al., 2012, Yates et al., 2021]. The present paper gives a distributional analogue. Age is costly because it induces geometric error under drift, and temporal roughness determines how quickly that cost accumulates.

Across these neighboring literatures, the closest theme is that nonstationarity makes retention scale matter. The distinguishing claim here is more specific: finite memory has a roughness-indexed temporal horizon, the worst-case cube-root law is the $H = 1$ member of that family, and temporal validity can fail before change becomes statistically detectable.

# 5 Limitations and Scope

**Path-class scope.** The main upper law is stated for deterministic path classes in which staleness can be controlled as a function of lag. The worst-case Lipschitz result is the $H = 1$ member of that family, while other roughness classes induce different horizon exponents.

**Lower-bound scope.** The structural floor is proved through a Gaussian location witness embedded in the ambient $W_2$-Lipschitz class. That is enough for a subclass-based minimax lower bound. The important distinction is between exponent-tightness, class-tightness, and constant-sharpness. At the Lipschitz endpoint $H = 1$, the Gaussian witness settles the first: it matches the cube-root scaling and shows that the horizon law is structural there. What remains open is the latter two questions for the full drift class, together with matching lower bounds for the broader Holder family $H \in (0, 1)$. The optimized ramp witness shows that the explicit lower-law constant can be improved substantially within the same argument, but the exact hardest path and the constant-sharp lower law for the full class are still open.

**Finite-sample carrier regime.** The clean cube-root law uses a carrier law of the form $C_K n^{-a}$, and the main text specializes to the root-$n$ case $a = 1/2$. That is the relevant low-dimensional Wasserstein regime and also the regime induced by fixed-$\varepsilon$ Sinkhorn surrogates on bounded domains. In the theorem statements this enters as a carrier-regime assumption for the non-identically distributed windowed sample, while lemma 2.5 records the i.i.d. reference case. Outside that carrier exponent, the same balancing argument gives a different horizon law, as noted in remark 2.8. Appendix A adds a triangular-array sanity check in a fixed-span regime, which supports this carrier interpretation empirically but does not replace a theorem. The main open technical question is therefore to identify natural conditions under which the windowed triangular array satisfies

$$\mathbb{E}\, W_2\left(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)}\right) \le C n^{-1/2},$$

or, failing that, to characterize the correct carrier exponent for the non-identically distributed window directly.

**Detectability versus validity.** The invalidity gap is an empirical consequence of the structural theory, not the primary theorem object. The paper shows that temporal validity can fail before a detector fires, but it does not claim a universal detection-delay theorem across all detector classes.

**What remains open.** Several questions remain deliberately open. The most immediate is the windowed triangular-array carrier theorem above. Beyond that, sharp constants in the horizon laws, matching lower bounds for broader Holder classes, online identification of the relevant roughness class, and transfer from deterministic path-class statements to stronger stochastic-process formulations all remain of interest. Those are natural next steps, but they are not prerequisites for the main structural claim. The present lower bound is already materially stronger than the conservative witness used in earlier drafts, but it should still be read as a structural floor rather than a final extremal characterization.

# 6 Conclusion

Under drift, memory is never purely beneficial. It reduces finite-sample noise, but it also ages. The resulting variance–staleness trade-off makes useful memory a finite-horizon object rather than an unlimited statistical resource.

The analysis shows that this horizon is roughness indexed. For path classes in which temporal misalignment grows like $\zeta n^H$, the useful-memory scale is

$$n_H^* \asymp (C_K/\zeta)^{2/(1+2H)}.$$

The cube-root law is therefore not the whole theory. It is the worst-case Lipschitz, $H = 1$, member of a broader family. The Gaussian location lower bound shows that the same cube-root scale is structural rather than an artifact of one particular estimator. The optimized ramp witness substantially strengthens the explicit lower-law constant, even though the result is still not constant-sharp.

The carrier law is written abstractly as $C_K n^{-a}$, and the main text specializes to the root-$n$ case $a = 1/2$. The Lipschitz upper law itself is stated with the correct $W_2^2$ convexity constant and with the carrier term treated as a regime assumption for the windowed triangular array rather than as a universal theorem.

The empirical section makes the same structure visible in finite samples. It reveals a structural U-curve, roughness-dependent scaling across path classes, a strictly positive invalidity gap between temporal validity and changepoint evidence, and measurable cost when the operating horizon departs from the useful-memory scale.

The paper therefore supports a simple conclusion. Useful memory has a finite horizon determined by how temporal roughness accumulates staleness, and retained evidence can become invalid before change becomes statistically detectable.

The main remaining technical gap is to justify the assumed carrier regime for the windowed triangular array in $W_2$ under natural conditions. Sharper constants, fuller lower-bound characterizations beyond the Gaussian witness, and online identification of the relevant roughness class come after that. Those questions refine the theory; they do not alter its main object.


# Acknowledgements

With gratitude to everyone who contributed ideas, feedback, tools, encouragement, and care along the way. This work stands on the generosity of colleagues, collaborators, open-source maintainers, and the broader research communities that made it possible.


# A Finite-Sample Geometry Check

For the finite-sample term in lemma 2.5, we estimate empirical $W_2$ between two i.i.d. samples from the same distribution using exact assignment in low and moderate dimension, then fit a log–log slope in $n$. We also probe the non-identically distributed window directly in one dimension by comparing a triangular-array sample, with one draw from each drifted component in the window, against an i.i.d. sample from the same window mixture. In that check the within-window drift span $\zeta n^H$ is held fixed, so the experiment targets the carrier-regime interpretation used in the main text rather than the large-span regime in which heterogeneity itself can become dominant. Finally, we check the Sinkhorn proxy at a few fixed values of $\varepsilon$.

The numerical check is only meant to support the modeling choice behind lemma 2.5 and remark 2.6. In low dimension, the raw i.i.d. Wasserstein term

Table 3: Empirical scaling check for the finite-sample term. The bounded-support i.i.d. $W_2$ experiment is close to the expected $n^{-1/2}$ rate in low dimension, the corresponding triangular-array check compares one sample from each drifted component against an i.i.d. sample from the window mixture, and the Sinkhorn proxy is reported at fixed $\varepsilon$. These rows are numerical support for the carrier-regime discussion rather than theorem statements.

| Setting | Estimated slope | Comment |
|---|---|---|
| $W_2$, $d = 1$ | -0.48 | close to $-1/2$ |
| $W_2$, $d = 2$ | -0.41 | slower decay |
| $W_2$, $d = 4$ | -0.25 | slower decay |
| $W_2$, $d = 5$ | -0.24 | slower decay |
| $W_2$, i.i.d. mixture, $d = 1$, $H = 0.5$ | -0.50 | fixed span $\zeta n^H = 0.25$ |
| $W_2$, triangular, $d = 1$, $H = 0.5$ | -0.45 | fixed span $\zeta n^H = 0.25$ |
| $W_2$, i.i.d. mixture, $d = 1$, $H = 1.0$ | -0.47 | fixed span $\zeta n^H = 0.25$ |
| $W_2$, triangular, $d = 1$, $H = 1.0$ | -0.41 | fixed span $\zeta n^H = 0.25$ |
| Sinkhorn $\varepsilon = 0.50$ | -0.21 | proxy constant changes |
| Sinkhorn $\varepsilon = 0.10$ | -0.53 | proxy constant changes |
| Sinkhorn $\varepsilon = 0.05$ | -0.67 | proxy constant changes |

is compatible with an $n^{-1/2}$ fluctuation scale, whereas higher-dimensional raw Wasserstein decays more slowly. In the one-dimensional fixed-span triangular-array check, the recovered slopes stay in the same rough vicinity as the corresponding i.i.d. mixture baseline, which is consistent with the carrier-regime reading used in the main text but does not amount to a theorem. For fixed $\varepsilon$, the Sinkhorn proxy acts as a measurement layer whose calibration constant changes with regularization and ambient geometry even when the horizon argument keeps the same formal shape. This empirical calibration is meant to sit alongside, not replace, the low-dimensional empirical-measure results of Fournier and Guillin [2015], Niles-Weed and Bach [2019], the fixed-$\varepsilon$ Sinkhorn rate of Genevay et al. [2019], and the recent nonstationary or triangular-array concentration literature referenced in remark 2.6.

# References

O. Deniz Akyildiz, Pierre Del Moral, and Joaquín Miguez. Gaussian entropic optimal transport: Schr"odinger bridges and the sinkhorn algorithm. *CoRR*, abs/2412.18432, 2024. doi: 10.48550/arXiv.2412.18432.

Arash A. Amini and Luciano Vinas. Wasserstein concentration of empirical measures for dependent data via the method of moments. *CoRR*, abs/2601.07228, 2026. doi: 10.48550/arXiv.2601.07228.

Omar Besbes, Yonatan Gur, and Assaf Zeevi. Non-stationary stochastic optimization. *Operations Research*, 63(5):1227–1244, 2015. doi: 10.1287/OPRE.2015.1408.

Albert Bifet and Ricard Gavaldà. Learning from time-changing data with adaptive windowing. In *Proceedings of the 2007 SIAM International Conference on Data Mining (SDM)*, pages 443–448, 2007. doi: 10.1137/1.9781611972771.42.

Wang Chi Cheung, David Simchi-Levi, and Ruihao Zhu. Learning to optimize under non-stationarity. In *Proceedings of the Twenty-Second International Conference on Artificial Intelligence and Statistics*, volume 89 of *Proceedings of Machine Learning Research*, pages 1079–1087. PMLR, 2019. URL https://proceedings.mlr.press/v89/cheung19b.html.

Rainer Dahlhaus. Fitting time series models to nonstationary processes. *The Annals of Statistics*, 25(1):1–37, 1997. doi: 10.1214/aos/1034276620.

Nicolas Fournier and Arnaud Guillin. On the rate of convergence in Wasserstein distance of the empirical measure. *Probability Theory and Related Fields*, 162 (3–4):707–738, 2015. doi: 10.1007/s00440-014-0583-7.

João Gama, Pedro Medas, Gladys Castillo, and Pedro Rodrigues. Learning with drift detection. In *Advances in Artificial Intelligence — SBIA 2004*, volume 3171 of *Lecture Notes in Computer Science*, pages 286–295. Springer, 2004. doi: 10.1007/978-3-540-28645-5_29.

João Gama, Indrė Žliobaitė, Albert Bifet, Mykola Pechenizkiy, and Abdelhamid Bouchachia. A survey on concept drift adaptation. *ACM Computing Surveys*, 46 (4):44:1–44:37, 2014. doi: 10.1145/2523813.

Aude Genevay, Lénaïc Chizat, Francis Bach, Marco Cuturi, and Gabriel Peyré. Sample complexity of Sinkhorn divergences. In *Proceedings of the 22nd International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 89 of *Proceedings of Machine Learning Research*, pages 1574–1583, 2019.

Ziv Goldfeld, Kengo Kato, Gabriel Rioux, and Ritwik Sadhu. Limit theorems for entropic optimal transport maps and the sinkhorn divergence. *Electronic Journal of Statistics*, 18(1):980–1041, 2024. doi: 10.1214/24-EJS2217.

Fabian Hinder, Valerie Vaquet, and Barbara Hammer. The window dilemma: Why concept drift detection is ill-posed, 2023. Preprint.

David V. Hinkley. Inference about the change-point in a sequence of random variables. *Biometrika*, 57(1):1–17, 1970. doi: 10.1093/biomet/57.1.1.

Andrew H. Jazwinski. *Stochastic Processes and Filtering Theory*, volume 64 of *Mathematics in Science and Engineering*. Academic Press, New York, 1970.

Jiashuo Jiang, Xiaocheng Li, and Jiawei Zhang. Online stochastic optimization with wasserstein based non-stationarity, 2020. URL https://arxiv.org/abs/2012.06961.

Rudolf E. Kalman. A new approach to linear filtering and prediction problems. *Journal of Basic Engineering*, 82(1):35–45, 1960. doi: 10.1115/1.3662552.

Sanjit Kaul, Roy Yates, and Marco Gruteser. Real-time status: How often should one update? In *Proceedings of IEEE INFOCOM*, pages 2731–2735, 2012. doi: 10.1109/INFCOM.2012.6195689.

Dominic S. T. Keehan, Edward J. Anderson, and Wolfram Wiesemann. Don't look back in anger: Wasserstein distributionally robust optimization with nonstationary data. *CoRR*, abs/2510.18566, 2025. doi: 10.48550/arXiv.2510.18566.

Alessio Mazzetto and Eli Upfal. Nonparametric density estimation under distribution drift. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 24251–24270. PMLR, 2023. URL https://proceedings.mlr.press/v202/mazzetto23a.html.

Jonathan Niles-Weed and Francis Bach. Sharp asymptotic and finite-sample rates of convergence of empirical measures in Wasserstein distance. *Bernoulli*, 25(4A): 2620–2648, 2019. doi: 10.3150/18-BEJ1077.

E. S. Page. Continuous inspection schemes. *Biometrika*, 41(1/2):100–115, 1954. doi: 10.1093/biomet/41.1-2.100.

Zhiyong Wang, Jize Xie, Yi Chen, John C. S. Lui, and Dongruo Zhou. Variance-dependent regret bounds for non-stationary linear bandits, 2024. URL https://arxiv.org/abs/2403.10732.

Roy D. Yates, Yin Sun, D. Richard Brown, Sanjit K. Kaul, Eytan Modiano, and Sennur Ulukus. Age of information: An introduction and survey. *IEEE Journal on Selected Areas in Communications*, 39(5):1183–1210, 2021. doi: 10.1109/JSAC.2021.3064980.

Jianjun Yuan and Andrew G. Lamperski. Trading-off static and dynamic regret in online least-squares and beyond. In *The Thirty-Fourth AAAI Conference on Artificial Intelligence*, pages 6712–6719. AAAI Press, 2020. doi: 10.1609/AAAI.V34I04.6149.